

lines are scanned on the screen in one second. The horizontal scan rate is controlled by the video card. Video cards are generally capable of generating higher scan rates but it depends on the speed at which the monitor is capable of scanning the electron beam across the screen and then returning it to the beginning of the next line. The monitors that had a higher horizontal scan rate were Sony CPD-E100 and ViewSonic E53 (30-70 KHz), whereas the others had a scan rate between 30-61 KHz.

The vertical scan rate on the other hand is the number of times the screen is displayed in one second and is popularly known as the refresh rate. Like the horizontal scan rate, it is always denoted as a range, but in Hertz, for example, 50-160 Hz. It is controlled by the vertical sync rate by the video card. Vertical scan rate is limited by the horizontal scan rate and the maximum resolution of the monitor.

Almost all the monitors had a refresh rate in the range of 50-120 Hz. Refresh rate is very subjective because at 60 Hz you might find the screen flickering whereas others might not find it so. So when buying a monitor, check out the range of horizontal and vertical refresh rate.

Video bandwidth: Normally, most of us tend not to notice this feature while buying a monitor as this is not mentioned anywhere on the box and may not be mentioned in the manual either. But it is an important feature as greater the video bandwidth of the monitor, higher the refresh rate it can take at a higher resolutions, thus affecting image quality. Video bandwidth is the product of refresh rate and maximum resolution (horizontal resolution x vertical resolution x refresh rate= video bandwidth) and is measured in mega hertz (MHz).

In the 15-inch category only ViewSonic E53 and Sony CPD-E100 had a video bandwidth of 110 MHz and 108 MHz, respectively, whereas the Samsung 55E had a bandwidth of 80 MHz, which is still on the higher side and gives the monitor the ability to work at 1024x768 at a refresh rate of 72 Hz .

Video bandwidth is higher in 19-inch monitors than in 15-inch monitors because of the screen size. The monitor that offered the best bandwidth was Samsung SyncMaster 955DF (240 MHz), while the Samsung 900NF had the lowest video bandwidth (135 MHz) in this category.

Among the 21-inch monitors, the LG Studioworks 221U had 30 MHz

TFT Display Tech

Flat Panel displays, also called Liquid Crystal displays (LCDs), have shown improvement in terms of features, technology and price by quantum leaps over the past year. In the past, they were accessible only to that select group of people who had the money and wanted to have the most appealing display systems available.

Two of the main technologies available in Flat Panel display systems are Active Matrix displays, more popularly known as TFTs (Thin Film Transistors), and Passive Matrix displays or DSTN (Dual Scan Twisted Nematic).

LCD screens use liquid crystals. Invented by James Fergason in 1970, liquid crystals are organic compounds that exhibit characteristics of both solids and liquids.

When light is passed through this nearly transparent substance, it gets aligned with the molecules of the liquid crystals as it passes through. Additionally, if an electric field is applied to these crystals, their alignment can be changed and consequently, so can the behavior of the light

passing through it. This works in much the same way as do blinds on a window—depending on the orientation of the slats, light is passed or blocked. In the case of the slats, we change their orientation by hand while in the case of TFTs, the microscopic orientation of the LCD elements are changed by passing a voltage between them.

In the construction of LCD panels, the liquid crystals are sandwiched between two very finely grooved layers that serve to arrange the crystals in accordance with the layers. Since the molecules follow the grooves, if the grooves are parallel, then so are the molecules. There are two grooved surfaces used where one set is aligned 90-degrees compared to the other, hence disallowing the passage of light. However, if an electric field is applied to one set of these grooves, these liquid crystals become aligned, allowing the passage of light. This principle is used in the creation of all types of Flat Panel screens used today, including DSTN and TFT screens.



TFT vs CRT Screens						
Display Type	Viewing Angle	Contrast Ratio	Response Time	Brightness	Power Consumption	Life
Passive Matrix LCD	40-100 degrees	40:1	300 ms	70-90	45 Watts	600,000 hours
Active Matrix LCD	> 140 degrees	140:1	25 ms	70-90	50 Watts	600,000 hours
CRT	> 190 degrees	300:1	NA	220-270	180 Watts	Years

Due to the technologies used, various types of display systems exhibit different performance and display characteristics. This is a brief comparison of the most common parameters between the CRT (Cathode Ray Tube), the Active Matrix LCD (TFT) and the Passive Matrix TFT (DSTN).